

Velocità
$v_{x,media} = \frac{\Delta x}{\Delta t} \quad (1a)$
$v_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt} \quad (1b)$

Accelerazione
$a_x = \frac{\Delta v}{\Delta t} \quad (2)$

Moto Rettilineo Uniforme
$x_f = x_i + vt \quad (3)$

Moto Uniformemente Accelerato
$v_{xf} = v_{xi} + a_x t \quad (4a)$
$x_f = x_i + v_{xi} t + \frac{1}{2} a_x t^2 \quad (4b)$
$v_f^2 = v_i^2 + 2a(x_f - x_i) \quad (4c)$

Moto Verticale
$t_c = \sqrt{\frac{2h}{g}} \quad (5a)$
$v_c = \sqrt{2gh} \quad (5b)$

Moto Armonico Semplice
$x(t) = A \sin(\omega t + \phi) \quad (6a)$
$v(t) = \omega A \cos(\omega t + \phi) \quad (6b)$
$a(t) = -\omega^2 A \sin(\omega t + \phi) \quad (6c)$
$T = \frac{2\pi}{\omega} \quad (7a)$
$f = \frac{1}{T} = \frac{\omega}{2\pi} \quad (7b)$
$\omega = \frac{v_t}{r} \quad (7c)$
$v_{max} = A\omega \quad (8)$

Moto Parabolico
$h = \frac{v_i^2 \sin^2 \theta_i}{2g} \quad (9a)$
$R = \frac{v_i^2 \sin(2\theta_i)}{g} \quad (9b)$
$y_f = \tan \theta_i \cdot x_f - \frac{g}{2v_i^2 \cos^2 \theta_i} x_f^2 \quad (9c)$

Moto Circolare Uniforme
$a_c = \frac{v_t^2}{r} = \omega^2 r \quad (10a)$
$v = \frac{2\pi r}{T} = \omega r \quad (10b)$
$\omega = \frac{v}{r} \quad (10c)$

Coordinate Moto Circolare
$\theta(t) = \theta_0 + \omega t \quad (11a)$
$x(t) = R \cos(\theta(t)) \quad (11b)$
$y(t) = R \sin(\theta(t)) \quad (11c)$

Moto Curvilineo
$\vec{a} = \vec{a}_{radiale} + \vec{a}_{tangenziale} \quad (12a)$
$a_r = \frac{v^2}{r} = \omega^2 R \quad (12b)$
$a_t = \frac{d \vec{v} }{dt} \quad (12c)$
$a = \sqrt{a_r^2 + a_t^2} \quad (12d)$

Seconda Legge di Newton
$\sum \vec{F} = m\vec{a} \quad (13)$
Unità: $N = \frac{kg \cdot m}{s^2}$

Forze di Attrito
$f_s \leq \mu_s n \quad (14a)$
$u_d < u_s \quad (14b)$

Piano Inclinato
$\sin(\theta) - \mu_d \cos(\theta) \geq 0 \quad \Rightarrow \quad \tan(\theta) \geq \mu_d \quad (15)$

Forza Elastico
$\vec{F} = -kx\hat{i} \quad (16a)$
$a = -\frac{k}{m}x = -\omega^2 x \quad (16b)$
$\omega = \sqrt{\frac{k}{m}} \quad (16c)$

Lavoro
$W = \vec{F} \cdot \vec{s} = Fs \cos \theta = F_T s \quad (17a)$
$W = \int_A^B F_T ds \quad (17b)$
Unità: $J = [N \cdot m] = [kg \cdot m^2 \cdot s^{-2}]$

Tipi di Lavoro
• Motore: $0 \leq \theta < \frac{\pi}{2}$ • Resistente: $\theta > \frac{\pi}{2}$ • Nullo: $\theta = \frac{\pi}{2}$

Potenza
$P_{media} = \frac{W}{\Delta t} \quad (18a)$
$P_{istantanea} = F_T v \quad (18b)$
Unità: $W = [J/s] = [kg \cdot m^2 \cdot s^{-3}]$

Energia Cinetica
$E_k = \frac{1}{2}mv^2 \quad (19)$

Teorema Energia Cinetica
$W = \Delta E_k = E_{k,f} - E_{k,i} \quad (20)$

Lavoro Forze
$W_{\text{peso}} = -(mgy_B - mgy_A) \tag{21a}$
$W_{\text{elastica}} = -\left(\frac{1}{2}kx_B^2 - \frac{1}{2}kx_A^2\right) \tag{21b}$
$W_{\text{attrito}} = -\mu_d N s_{AB} \tag{21c}$

Energia Potenziale
$E_{p,\text{molla}} = \frac{1}{2}kx^2 \tag{22a}$
$E_{p,\text{peso}} = mgy \tag{22b}$
Energia Meccanica
$E_m = E_K + E_P = \text{costante} \tag{23}$
Con attrito: $E_{m,B} = E_{m,A} - W_{\text{attrito}} \tag{24}$